New Mathematical Results from the Zeta / Random-Matrix / Prime-Gap Programme

Theorems, certified records, and exact laws that do not appear in the prior literature

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> Revision note (2 November 2026). This revision (i) applies the corrections from GPT-6 Astra's independent assessment of the 6 September version (`progress_assessment_2026_09_07.md` and `PRIME_CLAIM_AUDIT.md` in the companion repository): the "invisible to polynomial statistics of any degree" phrase in item 3 is narrowed to the protected algebra, Theorem 4.3 excludes the degenerate parameter k = N/2, the signed structure of the two-scale residual in Theorem 7.2 is made explicit, the pair-correlation LP value is labelled numerical rather than sharp, and the outward-rounded prime-gap certificates that were missing from the public tree are regenerated and committed; (ii) adds §3.7, the reviewed results of the 6 September push session (the 3-block limit theorem, the midpoint-insertion principle, the discriminant–force identity, the corrected ceiling conjecture on the AH support, and the barrier theorem for the last mile to an AH refutation). No status upgrade is made anywhere.

Abstract

This paper isolates, from a larger programme on Riemann zeros, random matrices and prime gaps, only those results that we believe to be new mathematics with independent value: statements that are proved (or machine-certified), that we have checked against the literature to the extent stated in §1, and whose content is not a restatement of a known theorem. Everything surveyed, standard, refuted, or merely numerical has been left out or is explicitly labelled as supporting evidence.

The results are:

1. Three unconditional prime-gap records. liminf(p_{n+2} − p_n) ≤ 173,438, liminf(p_{n+3} − p_n) ≤ 13,859,802, liminf(p_{n+4} − p_n) ≤ 1,120,662,828 (previous records 396,504 / 24,797,814 / 1,431,556,072). The new input is a variational theorem: an exact layer-cake reduction of the Maynard–Tao functional for product test functions to one-dimensional lower-tail probabilities, which recovers about a factor 3 in k over the closed-form bound in use since 2014. The H_2 and H_3 certificates are outward-rounded and use no probabilistic constant: M_{15,856} ≥ 8.00677 > 8 and M_{923,601} ≥ 12.00263 > 12 from Maynard's theorem and Bombieri–Vinogradov alone.
2. The signed-sieve no-gain theorem. With the decoding debt charged at face value, every signed Maynard–Tao weight is dominated by its own positive part; the "remove the square" enlargement of the variational problem is empty, so any gain from signed weights must be arithmetic.
3. A stopping time that separates CUE from Tao's ACUE across universality classes. For the finite backward heat flow on the characteristic polynomial, the first collision time D satisfies D ≥ −log cos(δ_min/2) ≥ δ_min²/8 for every configuration; N²D ≥ π²/8 for every non-clock ACUE configuration; and for Haar CUE, N^{8/3}D ⇒ G²/8 with P(G > x) = exp(−x³/72π), by two independent routes, one of which carries an explicit uniform stiffness tail P(S* > MN²) ≤ 4087·M^{−1/2}. Along the flow the two ensembles agree on every balanced moment of degree ≤ N (and, by Astra's generator theorem, on every protected trace moment under the forward flow), so the separation is invisible to the whole protected algebra of balanced trace moments of weight ≤ N, statically and dynamically. (Higher-weight statistics do see the lattice: on ACUE the 2N-th power trace satisfies |p_{2N}|² = N² identically, against E|p_{2N}|² = N for CUE; the point of the depth is that it is a first-passage time of the configuration, not a trace moment of any weight.)
4. The impostor fibre. Exact dimensions 0, 0, 2, 10, 80, 403, 1804 (N = 3, …, 9) of the space of measures on the ACUE support matching all CUE balanced moments of degree ≤ N; an explicit (N−3)-parameter family q = μ_ACUE·g(Σc mod N) with E g = 1, ĝ(±1) = 0; and Theorem E1: a chiral sub-family at total-variation distance ≈ 0.3 from ACUE that is invisible to every dihedral-invariant statistic, including the depth.

5. The Nyquist conservation law. For integer-marked configurations on $\mathbb{Z}/2N$ with mass N matching the finite CUE ramp $E|p_k|^2 = k$ for $1 \le k \le N-1$, all multiplicity is stored in the one unconstrained Fourier row: $E\,\Sigma\, m_x(m_x - 1) = (E|p_N|^2 - N)/2N$; a slack identity forces the simple-site fraction $\ge \tfrac{1}{2} + \csc^2(\pi/2N)/(2N^2) \to \tfrac{1}{2} + 2/\pi^2$ with equality exactly on {0,1,2}-hard-core configurations; and the density curve $\Phi(\rho) = 1 - \rho + \sin^2(\pi\rho)/(\pi^2\rho)$. Companion exact laws: the fourth-moment covariance $\mathrm{Cov}(|p_k|^2, |p_\ell|^2) = k^2 \cdot 1\{k=\ell\} - 2(k+\ell-N)_+$ of the finite Fermi sea, and the exact total-variation formula $d_{TV} = (2r)^{\wedge}g|\cos g\varphi - \cos g\psi|$ for magnetic-cycle determinantal laws that share all correlations of order $< g$.
6. Theorems about the Montgomery–Taylor method. A closed-form finite Galerkin ladder $q_n \to q^*$ with exact $n^{\wedge}\{-2\}$ error; a strictly stronger distinct-zero decoding $N_d \ge 0.8362503 \cdot N + p$; the exactly flat RH-independence frontier; the edge no-go (pointwise edge information at bandwidth one is worth exactly zero); weight freezing of derivative statistics; the non-vanishing third-moment defect $\Phi_3 = -0.0117753128$ at the Montgomery–Taylor window; and a correction to the record: the sharp bandwidth-one pair-correlation LP value is ≈ 0.679, not the 0.6818287 in circulation.
7. An explicit arithmetic refutation criterion for the alternative hypothesis (Astra): under RH and AH-pairs, a two-scale mean square of ζ'/ζ converges to $W_{AH} \in (0.06239, 0.06240)$, against the sine-kernel value 0.0822714…; hence $\liminf W_T \ge 1/16$ under RH refutes AH-pairs.
8. Addendum (push session, §3.7). The symmetric 3-block defect of the alternating clock has $\lim N^2 D = 2$ exactly, with its full asymptotic expansion; a closed-form midpoint-insertion principle for the heat flow of lattice defects; the discriminant–force identity $d/ds \log|\mathrm{disc}\, P_s| = -\Sigma V_j^2$ with log-concavity along the flow; the conjecture $N^2 D < 2$ on the AH support is false at $N = 12$ and is replaced by a domain-wall constant $C^* = 2.11466488\ldots$; and a barrier theorem: Astra's target is a linear functional of Montgomery's form factor, $W_T = \int K\, F_T + o(1)$, and every input the programme can prove is satisfied by the AH spectral measure, so the last mile is exactly one out-of-band value of F.

Each item states its hypotheses, its proof status, and what remains open. No famous conjecture is claimed.

1. Novelty protocol

A result is included only if (i) it is proved in the programme's files or machine-certified there, (ii) at least one independent adversarial review (a second model instructed to refute it) found no error in the final version, and (iii) we could not find it in the literature. For (iii) the honest position is this: during most of the programme no web access was available, and literature statements were recalled from training. On 6 September 2026 we ran targeted web searches for each headline item. What we found: the current H_1 records are 240 (Stadlmann, arXiv:2608.31126, August 2026), 236 and 212 (September 2026), and 186 (OpenAI, conditional on Kloosterman-sum axioms); none of these papers reports $m \ge 2$ records, and we found no $m \ge 2$ bound below the 2014/2023 values quoted above. We found no published result on first-collision times of the finite backward heat flow for CUE (the closest are Tao's 2017–2018 notes on heat flow and zeros, the Ben Arous–Bourgade extreme-gap law, and the Hall–Ho / Kabluchko heat-flow-conjecture line, none of which treats collision times). Rodgers–Vallabhaneni (2023) compute ACUE mixed moments in closed form but not the fibre of moment-matching measures. We found no occurrence of the density curve $\Phi(\rho)$ or the fourth-moment covariance formula. We found no treatment of signed Maynard–Tao weights with a decoding debt. This is evidence of novelty, not proof of it; Astra's own ledger records the novelty audit of the isolation lemma (§3.5) as incomplete because of its kinship with classical Lehmer-pair estimates.

Tags: [P] proved in the programme and reviewed; [C] machine-certified or exactly computed; [P/C] proved with a computed input; (cited) a published theorem used as input.

2. Bounded gaps between primes

2.1 Setting

Let M_k be the Maynard–Tao constant: the supremum of $\Sigma_{i=1}^{\wedge}k\, J_i(F)/I(F)$ over square-integrable F supported on the simplex $R_k = \{t_i \ge 0, \Sigma t_i \le 1\}$, where $I(F) = \int F^2$ and $J_i(F) = \int(\int F\, dt_i)^2\, dt_{-i}$. Maynard's theorem (cited; Ann. of Math. 181 (2015)): if the primes have level of distribution θ and $M_k > 2m/\theta$, then every admissible k-tuple has infinitely many translates containing at least m+1 primes. Bombieri–Vinogradov gives every $\theta < \tfrac{1}{2}$, so $M_k > 4m$ suffices.

2.2 The one-dimensional reduction and the layer-cake theorem [P]

Fix g: [0,∞) → [0,∞), piecewise linear on finitely many nodes, supported on [0,T], g ≢ 0. Put $c_2 = \int g^2$, $G(y) = \int_0^{\min(y,T)} g$ for y ≥ 0 and G(y) = 0 for y ≤ 0, and let X, X_1, X_2, … be i.i.d. with density g^2/c_2. Write $S_j = X_1 + \dots + X_j$. Take the product test function

$F(t) = \prod_{i=1}^k g(k\, t_i) \cdot 1[\Sigma\, t_i \le 1]$.

Lemma 2.1 (exact reduction). $I(F) = k^{-k}\, c_2^k\, P(S_k \le k)$ and $J^{(k)}(F) = k^{-(k+1)}\, c_2^{k-1}\, E[G((k - S_{k-1})_+)^2]$.

Proof. Substitute $x_i = k\, t_i$. Then $F^2 = \prod g(x_i)^2 \cdot 1[\Sigma x_i \le k]$ integrates to $c_2^k\, P(S_k \le k)$ times the Jacobian k^{-k}. For J, fix $x_1, \dots, x_{k-1}$ with sum σ; $\int F\, dt_k = k^{-1} \prod_{i<k} g(x_i) \cdot G((k-\sigma)_+)$, because g vanishes beyond T. Square and integrate. ∎

Theorem 2.2 (layer cake). $E[G((k - S_{k-1})_+)^2] = \int_0^T 2G(u)g(u) \cdot P(S_{k-1} < k - u)\, du$.

Proof. For y ≥ 0, $G(y)^2 = \int_0^y 2G(u)g(u)\, du$ since $(G^2)' = 2Gg$. Hence $G((k-S)_+)^2 = \int_0^T 2G(u)g(u) \cdot 1[u < k - S]\, du$, and Fubini gives the claim. ∎

Corollary 2.3 (certified lower bound). For any numbers $\beta_j \ge P(S_{k-1} \ge k - b_j)$ on a partition $\{[a_j, b_j]\}$ of [0,T],

$M_k \ge c_2^{-1}\, \Sigma_j \max(0, 1 - \beta_j) \cdot (G(b_j)^2 - G(a_j)^2)$,

since the true tail $P(S_{k-1} < k - u)$ is monotone in u, so on each cell it is ≥ 1 − β_j.

The point is that Corollary 2.3 needs only upper bounds on lower-tail probabilities of a sum of i.i.d. bounded variables, for which Chernoff (with the exponential majorised by chords on the piecewise-linear data, so that every integral is a rational number), a one-big-jump bound, or Berry–Esseen can be used, and any such family gives a valid certificate. This replaces the closed-form truncation bound used for every m ≥ 2 record since Polymath 8b (2014), whose deficit from log k was ≈ 2.3–2.9; the exact layer-cake accounting recovers ≈ 0.12 and shaped sub-exponential profiles $g = e^{-(t/T_1)^\kappa}/(1 + At)$ on a long support recover ≈ 0.49 more, together a factor ≈ 3 in k. The 2023–2026 improvements of H_1 upgraded the arithmetic input while keeping the crude variational bound, which is why this gain was available.

2.3 The records [P/C]

Theorem 2.4. $\liminf(p_{n+2} - p_n) \le 173{,}438$; $\liminf(p_{n+3} - p_n) \le 13{,}859{,}802$; $\liminf(p_{n+4} - p_n) \le 1{,}120{,}662{,}828$.

Certificates. With g the optimised 400-node profile (exact dyadic node values, sha256-pinned):

k	m	threshold	certified M_k (Berry–Esseen, C = 0.56)	certified M_k (no Berry–Esseen)	tuple diameter
15,856	2	8	8.013326752751306578...	8.006774080089994410774	173,438
923,601	3	12	12.006666706750...	12.002630349905 71191492	13,859,802
56,000,000	4	16	16.0655	—	1,120,662,828

The H_2 and H_3 certificates were recomputed independently in outward-rounded ball arithmetic (python-flint/arb, 200 bits, adverse endpoint at every inequality; cross-checked with mpmath interval arithmetic plus a 2^{-150} guard band; both agree to all 30 printed digits and reproduce the historical float bit-for-bit). The Berry–Esseen-free column uses Chernoff + chord + Markov only, so the H_2 and H_3 records depend on no probabilistic literature constant. Admissibility: for k = 15,856 the explicit symmetric tuple {−86,719, …, 86,719} misses a class modulo each of the 1,847 primes ≤ k (two independent implementations); for k = 923,601 a repaired Hensley–Richards tuple (73,001 primes checked, two implementations; the classical fallback of diameter 14,505,780 also verified); for $k = 56 \cdot 10^6$ the first $56 \cdot 10^6$ primes past k, with π-anchors checked against published values.

Artifacts (revision, 2 November 2026). The outward-rounded certificates and run logs cited in `overnight/fable/r1_h2_interval_cert.md` were written to a directory excluded by the repository's `.gitignore` and were therefore absent from the public tree at commit a64b698, as Astra's evidence audit noted. They have been regenerated from the committed script and force-added: `overnight/fable/data/h2_k15856_interval_cert.json`, `h3_k923601_interval_cert_Mk_only.json` and their run logs, reproducing the numbers above exactly (arb backend 1.3 s, mpmath interval backend

32 s; the k = 15,856 tuple admissibility check re-run: 15,856 distinct entries, diameter 173,438, 1,847 primes). Scope, as Astra's audit states: the new outward-rounded certificate covers H_2 fully and the M_k part of H_3; the H_3 tuple and the H_4 certificate rest on the August announcement's checks (two-implementation admissibility; the historical 200-bit arb run), not on the new script. These remain computer-assisted results awaiting external replication.

Status. Unconditional given Maynard's theorem and Bombieri–Vinogradov (both cited). The certificate chain is machine-checked but not formalised. An earlier audit (Astra) correctly noted that the committed certifier script was 50-digit floating point with a slack factor, not outward rounding; the re-certification above removes that objection.

2.4 The signed-sieve no-gain theorem [P]

Maynard–Tao weights are squares, $w(n) = (\Sigma_d \lambda_d)^2 \geq 0$. Positivity is what makes the decode valid: if $S_2 - mS_1 > 0$ with $S_1 = \Sigma\, w(n)$, $S_2 = \Sigma\, w(n)\nu(n)$, $\nu(n) = \#\{i : n + h_i \text{ prime}\}$, then some n has $\nu(n) \geq m+1$. For signed w one must pay the debt $D(w) = \Sigma_{w(n)<0} |w(n)|(m - \nu(n))_+$, after which $S_2 - mS_1 - D(w) > 0$ again implies the conclusion. Zhang's Landau–Siegel programme and Iwaniec's well-factorable $\lambda^{\pm}$ both suggest that signed weights are the natural next class.

Theorem 2.5. Write $w = w_+ - w_-$ with $w_+, w_- \geq 0$ of disjoint support. Then

$$S_2 - mS_1 - D(w) = \Sigma_n w_+(n)(\nu(n) - m) - \Sigma_n w_-(n)(\nu(n) - m)_+ \leq \Sigma_n w_+(n)(\nu(n) - m).$$

Proof. For every integer $\nu \geq 0$ and $m \geq 1$, $(\nu - m) + (m - \nu)_+ = (\nu - m)_+$: both sides equal $\nu - m$ if $\nu \geq m$ and 0 otherwise. Multiply by $w_-(n)$, sum, and subtract from $\Sigma\, w(n)(\nu(n) - m)$. ∎

So every signed weight is dominated, at face-value debt, by its own positive part: the variational problem does not enlarge. Positivity has a second job, boundedness ($\|w\|_1 = S_1$). Diagnostics on an exact finite model (n uniform on $\mathbb{Z}_W$, ν a coprimality count, weights in a level-L feature span, rational arithmetic) confirm that the apparent phase transition in the debt price β occurs at $\beta^* = 23051796480/10991046857 = 2.0973\ldots$ and is a normalisation artefact. The theorem says exactly where a signed route can live: (i) debt strictly below face value, which needs an exceptional character (Zhang), or (ii) a weight w that is evaluable while w_+ is not (well-factorable λ). Under a tuple-residue well-factorable estimate at level θ the conditional price list is $H_1 \leq 130, 114, 94, 80$ at $\theta = 4/7, 7/12, 3/5, 5/8$ [C].

3. The finite de Bruijn–Newman depth

3.1 Definition and dynamics [P]

For monic $P(z) = \prod_{j=1}^{N} (z - e^{i\theta_j}) = \Sigma\, a_j z^j$ define $P_s(z) = \Sigma\, a_j e^{s\cdot j(N-j)} z^j$ and the depth $D(P) = \inf\{s > 0 : \mathrm{disc}(P_s) = 0\}$. P_s stays self-inversive, so zeros stay on the circle until two collide; D is the first collision time. In terms of the real trigonometric polynomial $Q_0(x) = \prod \sin((x - \theta_j)/2)$, P_s corresponds (up to a scalar) to $e^{s\partial_x^2} Q_0$: the flow is the ordinary forward heat equation on Q_0, which is the direction that destroys real-rootedness. D is the finite analogue of $-\Lambda$ for the de Bruijn–Newman constant.

Lemma 3.1. Until the first collision, $d\theta_j/ds = -\Sigma_{k\neq j} \cot((\theta_j - \theta_k)/2)$.

Proof. $\partial_s P_s = (N D - D^2)P_s$ with $D = z\partial_z$. At a simple zero z_j, $\dot{z}_j = -(\partial_s P)(z_j)/P'(z_j) = -(N-1)z_j + z_j^2 P''(z_j)/P'(z_j) = -(N-1)z_j + 2z_j^2 \Sigma_{k\neq j}(z_j - z_k)^{-1}$. With $z = e^{i\theta}$ one has $2z_j/(z_j - z_k) = 1 - i\cot((\theta_j - \theta_k)/2)$; summing over $k \neq j$ the $(N-1)$ cancels and $\theta_j = \dot{z}_j/(iz_j)$ gives the claim. ∎

3.2 The comparison theorem and the lattice floor [P]

Lemma 3.2. $g' = -2\cot(g/2)$, $g(0) = g_0 \in (0, 2\pi)$ has the exact solution $\cos(g(s)/2) = e^s \cos(g_0/2)$, so g vanishes at $s = -\log\cos(g_0/2)$, and $-\log\cos(x/2) \geq x^2/8$ on $[0, \pi)$ with equality only at 0.

Theorem 3.3 (Theorem A). For every configuration and every adjacent pair with gap g, $g' \geq -2\cot(g/2)$ for all s before the first collision. Hence $D \geq -\log\cos(\delta_{\min}/2) \geq \delta_{\min}^2/8$.

Proof. For adjacent (a,b) with $g = \theta_a - \theta_b > 0$ across the short arc, Lemma 3.1 gives $g' = -2\cot(g/2) - \Sigma_{k\neq a,b}[\cot(x_a^k/2) - \cot(x_b^k/2)]$ with $x_j^k = (\theta_j - \theta_k) \bmod 2\pi \in (0,2\pi)$. Adjacency means $x_a^k = x_b^k + g$ with both in $(0, 2\pi)$, and $\cot(x/2)$ is strictly decreasing there, so each bracket is negative and enters with a plus sign. Cyclic order is preserved until the first collision, so the inequality holds for every gap for all $s < D$, and comparison with Lemma 3.2 started from each $g_i(0)$ shows no gap can vanish before $\min_i(-\log\cos(g_i(0)/2))$. ∎

Theorem 3.4 (Theorem C(i)). Every non-clock ACUE configuration has $\delta_{\min} = \pi/N$ exactly, hence $N^2 D \geq \pi^2/8 = 1.2337005501\ldots$.

Proof. The support is the 2N-th roots of unity, so every gap is a positive multiple of π/N and the N gaps sum to $2\pi = N\cdot(2\pi/N)$. If all gaps were $\geq 2\pi/N$ they would all equal $2\pi/N$, i.e. the clock. So some gap is exactly π/N, the smallest positive multiple. Apply Theorem 3.3. ∎

Exact enumeration of all 13,132 rotation orbits for $N \leq 10$ gives $N^2 D \in [1.31, 1.99]$ off the clock, $P(\text{clock}) = 2^{1-N}$ exactly, and the clock itself has $D = \infty$ (the polynomials $1 - cz^N$ are flow-invariant) [C]. The single-dislocation configuration (delete $e^{-i\pi/N}$ from the alternating clock, insert 1) has $N^2 D \to s^* = 1.419640342\ldots = 1.150717118\ldots\cdot\pi^2/8$, obtained independently from the lattice solver at $N = 20$ and from the first double zero of $G_s(u) = 2\cos(u/2) - 2\pi\int_0^{1/2} e^{s(1/4-y^2)}\cos((\pi+u)y)\,dy$, agreeing to $2\cdot 10^{-6}$ [C].

3.3 Invisibility to polynomial statistics [P]

Proposition 3.5 (frozen moments along the flow). Since P_s rescales a_j by the deterministic constant $e^{s\cdot j(N-j)}$, every polynomial in the coefficients $(a_j, \bar{a}_j)$ of balanced weight $\leq N$ evaluated on P_s is a polynomial of the same type evaluated on P. Because CUE and ACUE agree on all balanced moments of degree $\leq N$ (Tao), they agree on them for P_s at every s.

Theorem 3.6 (Astra; protected moments under the forward flow). Let $V_k = \sum_{j\neq k} \cot((\theta_k - \theta_j)/2)$, $L = \sum_k V_k \partial_{\theta_k}$ the generator of the repulsive circular Coulomb flow, and $p_m = \sum_k e^{im\theta_k}$. Then

$$L\,p_m = -m((N-m)p_m + \sum_{a=1}^{m-1} p_a\, p_{m-a}),$$

so the total positive Fourier weight of every term is exactly m, and for all N, all $1 \leq m \leq N$, all $r \geq 0$ and all forward times $t \geq 0$,

$$E_{\text{ACUE}}\, L^r|p_m|^2 = E_{\text{CUE}}\, L^r|p_m|^2,\quad E_{\text{ACUE}}|p_m(\Phi_t X)|^2 = E_{\text{CUE}}|p_m(\Phi_t X)|^2.$$

The proposal that iterated derivatives raise the degree of a low trace mode until lattice aliasing becomes visible is therefore false: no such derivative order exists. (Proof in `dynamic_generator.md`; the identity for $L\,p_m$ is checked symbolically and in cyclotomic arithmetic.)

Theorem 3.7 (Astra; an observable outside the protected algebra). With $D_{\text{force}} = \sum_i V_i^2$, $Q = \sum_{i\neq j} \csc^2((\theta_i - \theta_j)/2)$, $C_N = N(N^2-1)/3$: $D_{\text{force}} = Q - C_N$ identically, $E_{\text{CUE}}\, D_{\text{force}} = C_N$, $E_{\text{ACUE}}\, D_{\text{force}} = C_N/2$, $L\, D_{\text{force}} = -\sum_{i<j} \csc^2((\theta_i-\theta_j)/2)(V_i - V_j)^2 \leq 0$, $E_{\text{CUE}}\, L\, D_{\text{force}} = -\infty$, $E_{\text{ACUE}}\, L\, D_{\text{force}} = -2N(N^4-1)/15$.

Together, 3.5–3.7 make precise what the depth is escaping: the two ensembles agree on the whole algebra of protected trace moments, statically and dynamically; only rational observables of inverse gaps, and first-passage functionals, see the difference.

3.4 The matching upper bound: repaired stiffness and Theorem B′ [P]

Define, for an adjacent pair (a,b) with gap g and x_j^k as above, the stiffness

$$S^* := \sum_{k\neq a,b} \tfrac{1}{2}\cdot\max(\csc^2(x_b^k/2), \csc^2(x_a^k/2)) = \tfrac{1}{2}\sum_k \csc^2(\operatorname{dist}(\theta_k, \{\theta_a, \theta_b\})/2).$$

(An earlier version used only the b-endpoint; that bound is false in 34–49% of random configurations. The endpoint maximum is what the mean-value theorem actually gives, because $\csc^2(t/2)$ is decreasing on $(0,\pi]$ and increasing on $[\pi, 2\pi)$.) Exactly, $0 \leq$ background bracket $\leq g\cdot S^*$, and at the clock $S^* = (N^2-1)/6$.

Theorem 3.8 (Theorem B′). Suppose $S^*(s) \leq \Theta\cdot S^*(0)$ for $s \in [0,D) \cap [0, \delta^2/4]$, and let $\mu = \Theta\, S^*(0) + \kappa_0$ with $\kappa_0 \leq 4/\pi^2$ (the explicit constant in $-2\cot(g/2) \leq -4/g + \kappa(\delta/2)\cdot g$). If $\mu\delta^2 \leq 2$ then, whichever pair collides first,

$$-\log\cos(\delta/2) \leq D \leq -(2\mu)^{-1}\log(1 - \mu\delta^2/4) \leq (\delta^2/8)(1 + \mu\delta^2/4).$$

The window hypothesis holds with $\Theta = 2$ when all other gaps are $\geq 2\delta$, and under a one-sided counting hypothesis (H_C) $N_{ab}(\rho) \leq CN\rho + m_0$ with $CN\delta \leq 0.2071$; without such a hypothesis it fails (a 3-cluster with neighbour gap 1.01δ has $\sup S^*/S^*(0) = 9.5$). The proof is a linear differential inequality for g^2, integrated in closed form; no comparison lemma is needed.

3.5 The CUE background theorem and the depth law

Theorem 3.9 [P]. For CUE(N), all $N \geq 2$ and $L > 0$: $P(\delta_{\min} > L\cdot N^{-4/3}) \leq 4086/L^3$. For all $N \geq 3$ and $M \geq 1$: $P(S^* > M\cdot N^2) \leq 4087\cdot M^{-1/2}$, uniformly in N. First moment: $E[\#\text{gaps} \leq xN^{-4/3}] = (x^3/72\pi)(1 - N^{-2})(1 + O(x^2 N^{-2/3}))$.

Ingredients. The exact three-point function of CUE, $\rho_3 = (2\pi)^{-3}\prod_{i<j}|z_i - z_j|^2 \Sigma_{m_1<m_2<m_3}|s_\lambda(z)|^2$, gives the global clustering bound $\rho_3 \le C_3(N)\prod|z_i - z_j|^2$ with $C_3(N) = N^3(N^2-1)^2(N^2-4)/(69120\pi^3)$; a second-moment count of ordered triples with one pair at scale $LN^{-4/3}$ and a third point within c/N; and a layer-cake bound on the bulk sum. The tail exponent -3 in L is sharp; the exponent $\frac{1}{2}$ in M is not (the true tail is $\asymp M^{-5/2}$), because of Chebyshev on the bulk. The constant in the first statement was repaired after an independent review found a reversed inequality in one regime; the repair splits at the deterministic pigeonhole threshold $L = 2\pi N^{1/3}$ and is verified by script.

Theorem 3.10 (CUE depth law). For Haar CUE, $N^{8/3}D_N \Rightarrow G^2/8$ where $P(G > x) = \exp(-x^3/72\pi)$; equivalently $P(N^{8/3}D_N > t) \to \exp(-(2\sqrt{2}/9\pi)\cdot t^{3/2})$. Moreover $8D_N/\delta_{\min}^2 - 1 = O_P(N^{-2/3})$.

Two proofs. (i) Theorem 3.3 + Theorem 3.8 + Theorem 3.9 + the Ben Arous–Bourgade law (cited; Ann. Probab. 2013), which is complete modulo a short-time stability statement that S^* at $s = 0$ controls $\sup_{s<D} S^*(s)$ (stated, not proved). (ii) Astra's isolation lemma: for a sequence of configurations with smallest gap δ, endpoints at $\pm\delta/2$, and $B = \frac{1}{4}\Sigma_{k\neq\pm}\csc^2(\theta_k/2)$, if $\delta \to 0$ and $\delta^2 B \to 0$ then $D = (\delta^2/8)(1 + o(1))$, with explicit absolute constants when $\delta^2(B+1) \le \varepsilon_0$. It is proved directly on the scalar heat equation after an exact Galilean transformation that removes the common background drift, assumes nothing about the evolving background, and does not assume the closest pair collides first; the CUE input $E\,\Sigma_{\text{gaps} \le \varepsilon} B_i \le N^6\varepsilon^3/18$ gives $B_{\min}/N^2 = O_P(1)$. Route (ii) has two internal reviews and a quantified second-pass audit, and closes the localisation gap modulo the cited gap law. Neither route is Lean-checked or externally refereed.

Corollary 3.11 (universality class). For CβE with $\beta \in (0,\infty)$, the same argument with Feng–Wei's extreme-gap law (cited) gives $D_N = (\delta_{\min}^2/8)(1 + O_P(N^{-2/(\beta+1)+o(1)})) \asymp N^{-2-2/(\beta+1)}$, modulo one near-diagonal three-point density hypothesis (U3) verified so far only at $n = 2$, $\beta = 2$. The predicted rate is confirmed at three points: fitted exponents -1.012 / -0.710 / -0.501 for $\beta = 1, 2, 4$ against -1 / $-2/3$ / $-2/5$ [C]. The lattice is the singular endpoint $\beta = \infty$: $N^2\delta_{\min}^2 = \pi^2$ does not vanish, so $\rho \neq 1$ there.

What this says. CUE and ACUE agree on every polynomial statistic of degree $\le N$, statically and along the flow, and yet their depths lie in different universality classes, $N^{-8/3}$ versus N^{-2}. The mechanism is extreme-value: CUE admits rare pairs a factor $N^{-1/3}$ below the mean spacing and is always within $N^{-8/3}$ of losing real-rootedness, while the lattice quantises every gap at π/N. The alternative hypothesis fails this test by being too stable.

3.6 The Lagarias–Rodgers exchange rate and the threshold gap [P/C]

Let μ be the Lagarias–Rodgers hard-core extremum for bandwidth-one mimickers of the sine process ($\mu \ge \frac{1}{2}$ by the shifted half-lattice; they suggest, but do not prove, $\mu \le 0.606894$), and $\mu_\Lambda := \sup$ over mimickers of $\liminf N^2D$.

Proposition 3.12. A hard core of c mean spacings gives $\delta_{\min} = 2\pi c/N$, so $N^2D = \rho\cdot\pi^2c^2/2$ with $\rho \ge 1$, hence $\mu_\Lambda \ge \pi^2\mu^2/2$ and $\mu \le \sqrt{(2\mu_\Lambda)}/\pi$. In particular AH ($c = \frac{1}{2}$) forces $\liminf N^2D \ge \pi^2/8$, and a depth bound of 1.29 would give $\mu \le 0.511$.

Proposition 3.13 (the threshold gap). A single gap of λ mean spacings in a clock background breaks the floor $N^2D < \pi^2/8$ iff $\lambda < \lambda^* = 0.4719538$ (exact N-body dynamics, N-independent to 10^{-8}; rigid one-dimensional quadrature 0.4718999; expansion $\rho(\lambda) = 1 + (\pi^2/4 - 2)\lambda^2 + \ldots$). In a random CUE background $\lambda^* \approx 0.47$ with narrow spread. So reaching the floor through gaps needs gaps six percent below one half; the depth adds an explicit dressing factor to the hard-core statement, but no shortcut.

3.7 Addendum: exact constants on the AH support, and a barrier for the last mile (push session of 6 September, reviewed; revised 2 November)

A further proposer–refuter–repair round (files `overnight/fable/push/push_A…D` and their `refute_*` reports) produced the following.

Theorem 3.14 (the 3-block limit) [P]. Insert one extra root at a midpoint of the alternating clock (gap pattern $1,1,2,\ldots,2$ in units π/N, with the compensating deficit placed diametrically opposite, in either of its two placements). Then $\lim_{N\to\infty} N^2D_N = 2$ exactly. For the two mirror-symmetric placements the full expansion is $\tau_N := N^2D_N = 2 - 4/(3N^2) + c_2/N^4 + O(N^{-6})$ with $c_2 = -8/5$ and $-8/5 - \pi^2$ [P], and D_N is exactly the first zero of $P'_s(1) = \Sigma_j\, j\, a_j\, e^{s\,j(N-j)}$ [P, for every N]; for the asymmetric placement $\tau_N = 2 - 4/(3N^2) - 2.703\cdot N^{-8/3} + \ldots$, a cusp with coefficient $2(\pi^2/4)^{1/3}$ [C, confirmed to four digits at N up to 4096]. Proof idea. $Q_0 = 2^{1-N}\cos(Nx/2)\cdot\tan(x/2)$ in closed form; the heat flow of a trigonometric

polynomial is a Gaussian convolution, giving the local limit $q_\tau(u) = e^{-\tau/4}(u\cos(u/2) - \tau\sin(u/2))$ with an explicit $O(N^{-2})$ error in C^1; its zeros solve $u\cot(u/2) = \tau$, and since $u\cot(u/2) \le 2$ with equality only at $u = 0$, all zeros are real and simple for $\tau < 2$ and merge in a triple collision exactly at $\tau = 2$; Theorem A confines every other gap until $\tau = \pi^2/2 > 2$.

Proposition 3.15 (midpoint-insertion principle) [P]. For any real polynomial p and $L \in \{\cos(u/2), \sin(u/2)\}$, $e^{\tau\partial^2}[p\cdot L] = e^{-\tau/4}\cdot\mathrm{Re}$ (resp. Im)$[e^{iu/2}(e^{\tau\partial^2}p)(u + i\tau)]$. Consequences: the 5-block (two inserted midpoints) and two separated 3-blocks also give exactly 2 (a double zero at the block edge at $\tau = 2$); longer runs give larger constants — 7-run 2.0381, 9-run 2.0574, 11-run 2.0689 [C, two independent solvers] — so the 3-block is not the slowest-collapsing defect.

Theorem 3.16 (discriminant–force identity) [P]. With $V_j = \Sigma_{k\neq j}\cot((\theta_j-\theta_k)/2)$, along the flow

$d/ds \log|\mathrm{disc}\, P_s| = -\Sigma_j V_j^2 = N(N^2-1)/3 - \Sigma_{i\neq j}\csc^2((\theta_i-\theta_j)/2)$, $d/ds\, \Sigma_j V_j^2 = \Sigma_{i<j}\csc^2((\theta_i-\theta_j)/2)(V_i - V_j)^2 \ge 0$.

So $|\mathrm{disc}\, P_s|$ is strictly decreasing and log-concave in s off the clock; the flow is gradient ascent of the convex circular log-gas energy; and at $s = 0$, $E_{CUE}[d/ds \log|\mathrm{disc}|] = -N(N^2-1)/3$ while $E_{ACUE} = -N(N^2-1)/6$ (exact on all 160,000 ACUE orbits for $N \le 12$). Corollary: $D \ge C_N^{-1}\log(1 + C_N/A_0)$ with $A_0 = \Sigma_{i\neq j}\csc^2 - C_N$, exact at $N = 2$ but a factor $\asymp N$ weaker than Theorem A on the lattice. The midpoint family has the closed form $|\mathrm{disc}\, P_s| = 4(\lambda(N-1))^N[\sin(N\varphi/2) - \lambda\sin((N-2)\varphi/2)]^2/(\lambda^2-1)$, $\lambda = e^{s(N-1)}$, whose first collision at $\lambda = N/(N-2)$ gives $D = (2/M)\mathrm{artanh}(1/M)$, $M = N-1$, a second proof that $M^2D \to 2$.

The ceiling on the AH support [C/O]. The natural conjecture "$N^2D < 2$ for every non-clock ACUE configuration" is false: at $N = 12$ the configuration $[1^7,2,2,9,2,2]$ has $N^2D = 2.0000177$ (enumeration, 50-digit polyroots, and the N-body ODE agree). For $N \ge 11$ the maximiser is a run|clock|hole configuration $[1^{L-1}, 2^k, L+1, 2^k]$ with $L \approx N/2$, whose value increases with N: 2.0708 (32), 2.0926 (64), 2.1036 (128), 2.1091 (256), 2.1119 (512). The run constants τ^*_L increase to a domain-wall constant $C^* = 2.1146648843\ldots$, defined as the first double-zero time of the heat flow of $q_0(u) = \cos(u/2)/\Gamma(u/2\pi)$ (semi-infinite run joined to a semi-infinite clock), evaluated through a Hankel integral and matched to seven digits by the lattice extrapolation $\tau^*_L = C^* - 0.45/L$. Corrected conjecture: $\pi^2/8 \le N^2D < C^*$ on the whole AH support, approached from below. Proved sub-case [P]: if exactly one gap equals π/N then $N^2D \le \tau_1(N)$, an explicit comparison-ODE constant decreasing to $1.65737 < 2$. The ceiling file did not receive its own refuter pass (credit exhaustion); the 9- and 11-run constants and the $N = 12$ value were independently confirmed by the other refuters. The Nyquist row $|p_N|^2$ does not control the depth (correlation ≈ 0.13 across all orbits, no monotone envelope).

Theorem 3.17 (barrier for the last mile) [P, under RH; attribution corrected in review]. Astra's target of §7 is a linear functional of Montgomery's form factor:

$W_T = \int K(\alpha)\, F_T(\alpha)\, d\alpha + o(1)$, $K(\alpha) = \sinh(2)e^{-2|\alpha|} - \sinh(1)e^{-|\alpha|}$,

with $K > 0$ for $|\alpha| < \alpha_0 = \log(2\cosh 1) = 1.1269$ and $K < 0$ beyond (the kernel reproduces $W_{GUE} = 0.0822714431$ and $W_{AH} = 0.0623924180$ to 30 digits from $F_{GUE} = \min(|\alpha|,1)$ and $F_{AH} = \mathrm{dist}(\alpha,2\mathbb{Z}) + \Sigma_{m\neq 0}\delta_{2m}$). The band $|\alpha| \le 1$ contributes exactly the constant $B = 0.4560940$ of Theorem 7.2. F_{AH} satisfies Montgomery's theorem on the band, positivity of F and of the pair measure, and the Goldston–Gonek–Özlük–Snyder floor $F \ge 3/2 - |\alpha|$ on $(1, 3/2)$ (a GRH result, not RH-only, as the refuter established). Hence no derivation from these inputs, nor from band-limited correlations of any order or from integrality of multiplicities, can give $\liminf W_T \ge 1/16$, because F_{AH} is a model of all of them with $W < 1/16$. Quantitatively, the pair-positivity LP relaxation of the known inputs gives only ≈ 0.021 (≈ 0.024 with the GRH floor), uncertified; AH beats GUE in W solely through its even-integer Bragg atoms (contribution $e^{-2} - e^{-1} = -0.2325$); the missing information is one upper bound, $\limsup 2\int_{\alpha_0}^{\infty}|K|F_T \le 0.3936$, i.e. the $|K|$-average of F over $(1.127, \infty)$ must be ≤ 1.034 (AH has 1.051, GUE 1.000); and any hypothesis "$F \le 1$ in the measure sense on (α_0, ∞)" already gives $W \ge 0.0753 > 1/16$. The depth route does not evade this: Level B implies $\mu < \tfrac{1}{2}$ under (NR) (Theorem A′), needs a gap below $\lambda^* \approx 0.472$ to be produced by a single gap, and is invisible to every band-limited statistic. This is Tao's 2019 remark made quantitative: one value of F at a single $\alpha \in (1,2)$ with $\alpha \notin 2\mathbb{Z}+1$ refutes AH-pairs, and that value is the arithmetic input the programme does not have.

4. The impostor fibre [P/C]

Theorem 4.1 (dimension). The affine dimension of the set of signed measures on the ACUE support (N-subsets of the 2N-th roots of unity) matching every balanced moment of degree $\le N$ of CUE is 0, 0, 2, 10, 80, 403, 1804 for $N = 3, \ldots, 9$ (exact arithmetic; a tighter-tolerance recount at $N = 8$ gave 399, so the $N = 8$

entry is not settled and should be read as 403 or 399 pending an exact recount). Genuinely positive members exist from N = 5 on.

Theorem 4.2 (centre-of-mass family). For g: $\mathbb{Z}/N \to \mathbb{R}$, the measure $q_g(C) = \mu_{ACUE}(C)\cdot g(X(C))$, $X(C) = \Sigma_{c\in C} c \bmod N$, matches every balanced moment of degree $\le N$ if and only if $E_{ACUE}[g(X)] = 1$ and $\hat{g}(\pm 1) = 0$. This is an explicit $(N-3)$-parameter family at every N; X is exactly uniform under ACUE, and the coupling of a degree-$\le N$ balanced moment to a centre-of-mass frequency in $\{2, \ldots, N-2\}$ vanishes by a transport-cost count (hop budget $2N$ against cost $\ge 2N+2$). Together with the determinant-character (secant) and parity families this is one family, tilts by functions of $\det(U_C)$: on the lattice the global rotation is quantised and $\det^2$ becomes a shape invariant. Verified at $N = 5, \ldots, 8$ with worst moment error 10^{-12}.

Theorem 4.3 (E1, chiral blindness). For $N \ge 5$ and $2 \le k \le N-2$ with $k \ne N/2$ (at $k = N/2$ the tilt is $\sin(\pi X) \equiv 0$ and $q_{k,\varepsilon} = \mu_{ACUE}$; this exception, pointed out by Astra for $N = 6$, $k = 3$, was missing from the first version), the measures $q_{k,\varepsilon} = \mu_{ACUE}\cdot(1 + \varepsilon\cdot \mathrm{Im}\det(U_C)^{2k})$, $|\varepsilon| < 1$, are positive, lie in the CUE-moment fibre, have total-variation distance 0.22–0.30 from ACUE at $\varepsilon = 0.9$, and give exactly the same law as ACUE to every dihedral-invariant statistic: the depth, its clock atom, every pattern count, every Haar-lifted even marked-depth moment. The chiral tangent space of the fibre has dimension 1, 3, 39, 186 at $N = 5, 6, 7, 8$. On the reflection-symmetric sub-fibre the depth law is injective for $N \le 8$.

Proof idea. $\mathrm{Im}\det^{2k}$ is odd under reflection $C \mapsto -C$ while μ_{ACUE} and every dihedral-invariant functional are even, so the tilt integrates to zero against any such functional; balanced moments of degree $\le N$ are killed because $\det(U_C)^{2k} = e^{2\pi i k X/N}$ is a pure centre-of-mass mode of frequency $k \in \{2, \ldots, N-2\}$, exactly the range that Theorem 4.2's transport-cost argument decouples from every balanced moment of degree $\le N$ ($\hat{g}(\pm 1) = 0$ and $E\, g = 1$ hold for $g = 1 + \varepsilon \sin(2\pi k X/N)$). ∎

Consequence for zeta. The functional equation's reflection symmetry kills exactly this chiral half at zero cost (39 of 80 directions at $N = 7$) and provably cannot do more. Any statistic proposed to refute AH must be checked against the reflection-symmetric fibre, where the depth is already injective at small N.

5. The Nyquist conservation law and companion exact laws

5.1 The setting

Let $m = (m_x)_{x\in\mathbb{Z}/2N}$ be random nonnegative integer marks with $\Sigma m_x = N$ in every sample, and $p_k = \Sigma_x m_x e^{i\pi k x/N}$. The reference law is the rank-N consecutive-band projection determinantal process on $\mathbb{Z}/2N$ (the finite Fermi sea, whose scaling limit is ACUE and the half-lattice alternative hypothesis), with form factor $E|p_k|^2 = \min(k, N)$. Impose only the open rows $E|p_k|^2 = k$ for $1 \le k \le N-1$, leaving the Nyquist row $k = N$ free.

5.2 Three exact identities [P]

Theorem 5.1 (collisions live only at Nyquist). For every such law, $E\, \Sigma_x m_x(m_x - 1) = (E|p_N|^2 - N)/(2N)$.

Proof. Parseval on $\mathbb{Z}/2N$: $\Sigma_{k=0}^{2N-1}|p_k|^2 = 2N\, \Sigma_x m_x^2$. Now $p_0 = N$, $|p_{2N-k}| = |p_k|$, and the open rows give $\Sigma_{k=1}^{N-1} E|p_k|^2 = N(N-1)/2$, with the mirrored rows $k = N+1, \ldots, 2N-1$ contributing the same. So $2N\, E\, \Sigma\, m_x^2 = N^2 + N(N-1) + E|p_N|^2$, i.e. $E\, \Sigma\, m_x(m_x - 1) = E\, \Sigma\, m_x^2 - N = (E|p_N|^2 - N)/(2N)$. ∎

Consequently, closing the Nyquist row $E|p_N|^2 = N$ forces all marks into $\{0,1\}$: the complete finite ramp admits only simple configurations, and the one unmeasured Fourier row is exactly where all multiplicity is stored.

Theorem 5.2 (slack identity). Let $s_1 = \#\{x : m_x = 1\}$ and $h(j) = 0$ for $j \le 2$, $h(j) = j(j-2)$ for $j \ge 3$. Then, configuration by configuration,

$$s_1 - \Sigma_x m_x m_{x+1} - \Sigma_x h(m_x) = 2N - (1/2N)\, \Sigma_k |p_k|^2(1 + \cos(\pi k/N)),$$

and the Nyquist coefficient $1 + \cos\pi = 0$ makes the left side blind to the free row. Taking expectations under the open ramp and using $2\Sigma_{k<N} k\cos(\pi k/N) = N - \csc^2(\pi/2N)$,

$$E\, s_1 - (N/2 + \csc^2(\pi/2N)/(2N)) = E\, \Sigma_x m_x m_{x+1} + E\, \Sigma_x h(m_x) \ge 0,$$

with equality exactly on hard-core $\{0,1,2\}$-configurations (no adjacent occupied sites, no mark above 2).

Corollary 5.3 (density curve). $E\, s_1/N \ge \tfrac{1}{2} + \csc^2(\pi/2N)/(2N^2) \to \tfrac{1}{2} + 2/\pi^2 = 0.702642367284\ldots$; and matching the rank-L band process at every row except Nyquist forces $E\, s_1/L \ge \Phi(\rho) := 1 - \rho + \sin^2(\pi\rho)/(\pi^2\rho)$ in the limit $L/M \to \rho$. At low density $\Phi(\rho) = 1 - (\pi^2/3)\rho^3 + O(\rho^5)$: the sine kernel's quadratic repulsion becomes a

cubic collision-rigidity loss.

The value ½ + 2/π² is not an improvement of δ_MT = 0.6725 and neither bounds the other: the first is a simple-site fraction in a marked integer model with an open Nyquist row, the second a proportion of simple critical zeros proved from arithmetic. Their kinship is that in both, a low-order spectral certificate is strengthened by integrality and a missing Fourier direction marks the boundary of what the certificate can see. The realisation of the equality face at all sizes is open; the series reduces it to a three-tile integral language with exact no-go theorems for every natural construction, and an explicit (M, L) = (6, 3) law shows that realising the extremal pair function does not imply attaining the simple-site equality (the mark-≥3 channel).

5.3 Fourth moments of the finite Fermi sea [P/C]

Theorem 5.4. For the half-filled finite Fermi sea on $\mathbb{Z}/2N$ and $V_k = |p_k|^2$, $1 \le k < N$,

$$E[V_k V_\ell] = k\ell + k^2 \cdot 1\{k=\ell\} - 2(k + \ell - N)_+, \quad \mathrm{Cov}(V_k, V_\ell) = k^2 \cdot 1\{k=\ell\} - 2(k + \ell - N)_+,$$

and the covariance matrix has smallest eigenvalue exactly 1. (Fermionic normal ordering: the diagonal k^2 is the exchange term, the negative correction counts the two cyclic wrap intervals when two particle-hole excitations of total momentum $k+\ell$ exceed the band; verified by exact enumeration of all C(2N,N) atoms for $N \le 6$ to 10^{-9}, $\lambda_{\min} = 1$ at every tested N.) Consequences: the CUE target $(1, \ldots, N-1)$ is interior to the convex hull of binary feature rows with inradius $\ge 1/(N^2 - N/2)$; and a repair theorem: a candidate law on the hard-core face matching the ramp to feature error ε_N can be mixed with an explicit binary correction law to match the ramp exactly at simple-site cost $\le N\,\varepsilon_N/(\rho_N + \varepsilon_N)$, so feature error $o(N^{-2})$ suffices for o(N) slack.

5.4 An exact total-variation law for holonomy [P/C]

Theorem 5.5. On the g-cycle with magnetic kernel $K_\varphi = aI + rH_\varphi$ (forward edge phase $e^{i\varphi}$, $0 < 2r < \min(a, 1-a)$), every principal minor of order $< g$ is independent of φ, $\det K_\varphi - \det K_\psi = 2(-1)^{g-1} r^g (\cos g\varphi - \cos g\psi)$, and

$$d_{TV}(P_\varphi, P_\psi) = (2r)^g\, |\cos g\varphi - \cos g\psi|.$$

The connected stationary version keeps all correlations of order $< g$ equal while $d_{TV} \to 1$. This is the cleanest closed-form statement we know that bounded-order correlations cannot identify a determinantal law, with the entire information loss computed; orientation (the sign of the holonomy) is invisible to all sampling statistics. Companion exact witnesses: projection DPPs on $\mathbb{Z}/12$ with Fourier supports {0,1,4,6} and {0,1,3,7} have identical one- and two-point functions but triple probabilities $(8 \mp \sqrt{3})/432$; marked configurations u = (0,0,0,0,0,2,0,0,0,0,0,2,2), v = (0,0,0,0,1,1,0,0,0,0,1,3) on $\mathbb{Z}/12$ have identical complete autocorrelation yet simple-site counts 0 and 3, lifting to aperiodic model sets with equal pure-point diffraction and simple-particle fractions 0 and ½.

6. Theorems about the Montgomery–Taylor method

The 2026 Lean-verified theorem (Anthropic) proves unconditionally that at least $\delta_{MT} = 3/2 - (1/\sqrt{2})\cot(1/\sqrt{2}) = 0.672500703679\ldots$ of the zeros are simple and on the line, by measuring finite compressions of the indefinite Weil form from the prime side and decoding positive inertia through rank–trace inequalities. The following are new statements about that method, read at the Lean source.

Theorem 6.1 (Galerkin closed form) [P/C]. Partition $[-½, ½]$ into n cells and minimise the exactly integrated piecewise-constant energy $q(v) = \int v^2 + \iint |s-t|\,v(s)v(t)$. With $a_n = 1 + 1/(3n^2)$ and $\theta_n = \arccos(1 - 1/(n^2 a_n))$, the minimum is exactly $q_n = ½ + (a_n\, n/2)\cdot \sin\theta_n \cdot \cot(n\theta_n/2)$, and $2 - q_n = \delta_{MT} - [\csc^2(1/\sqrt{2}) - \sqrt{2}\cot(1/\sqrt{2})]/(24n^2) + O(n^{-4})$. (Agreement with direct quadratic programming $4\cdot 10^{-14}$ for $n \le 80$.) This converts 0.6725007… from the output of an optimisation run into a checkable finite ladder.

Theorem 6.2 (distinct-zero decoding) [P]. Combining the c = 2 inequality of the method with the zero-count identity once (not twice) gives $N_d \ge 0.8362503\cdot N + p$, where N_d counts distinct zeros and p off-line pairs: slope +1 in the off-line pair density, strictly stronger than the c = 3 decoding in the repository (slope ½). A five-line Lean edit.

Theorem 6.3 (flat frontier) [P]. The certified simple-zero proportion as a function of the off-line pair density π is $\delta(\pi) \equiv 0.6725007$ for all $\pi \in [0, 0.16375]$: assuming RH buys nothing within this method, because the binding adversary is on-line double zeros (exact 4p-budget/mass cancellation). At maximal off-line density all zeros are forced simple. Corollary: the density of off-line reflection pairs is at most $(1 - \delta_{MT})/2 =$

0.163749648160….

Theorem 6.4 (edge no-go) [P]. Every admissible bandwidth-one window has $\hat{r}(\pm 1) = 0$ (Fejér–Riesz), so the pointwise edge hypothesis $|F(1) - 1| \le \varepsilon$ certifies exactly 0.6725 even at $\varepsilon = 0$. The finite Nyquist conservation law of §5 is powered by lattice aliasing that $\mathbb{R}$ lacks; the correct zeta-side edge object is the Cesàro mean $F(\infty,T)$, with the exact collision identity $C(T) = N^{*}\cdot F(\infty,T) - N(T)$.

Theorem 6.5 (weight freezing) [P]. Derivative power sums are weight-homogeneous in the original power sums, so all bandwidth-one holomorphic statistics of the derivative process (the ξ' analogue) are frozen on the impostor fibre; "differentiate then pair-correlate at bandwidth one" cannot work. AH is not closed under differentiation (exact $N = 3$ counterexample), and on the circle the stationary points of Z are the roots of $zZ' - (N/2)Z$, all on the circle.

Theorem 6.6 (rigidity of the inertia lemma) [P]. An exact seven-term deficiency decomposition turns the method's Lemma R into an identity; an ε-near-equality configuration lies within Frobenius distance $\sqrt{\varepsilon}$ of an exact equality configuration. Zeta-side: the total deficiency over Montgomery–Taylor windows is $\le (q^{*} - 1)\cdot N(T) = 0.3275\cdot N(T)$, and on-line pairs with taper overlap $\ge \tau$ number $\le 0.1637\cdot N(T)/\tau^{2}$.

Proposition 6.7 (the third moment is informative at the MT window) [P/C]. At the flat window the sine-Gram data satisfies $m_3 - 3m_2 + 2m_1 = 0$ identically (a new rigidity fact holding at every finite N for ACUE), which is why $\operatorname{tr} Q^{3}$ looked worthless; at the Montgomery–Taylor window itself $\Phi_3 = m_3 - 3m_2 + 2m_1 = -0.0117753128$ (closed trigonometric form, verified at 30 digits). The obstruction to monetising it is precisely one missing lemma, $\|(c^{-1}\hat{A})_{-}\| \le M_{-}$ uniform in T, which would give 0.6796896 ($M_{-} = 2$) up to 0.6844924 ($M_{-} \le 1$); window engineering provably cannot substitute (the clump term forces the row-sum cap $M = A_0 \log T$ for every window).

Proposition 6.8 (the bandwidth-one pair-correlation LP: a numerical value) [C, correction to the record]. The value 0.6818287 carried in earlier files as "the pair-correlation ceiling" is a specific stability-inequality witness, not the sharp value of the $F \ge 0$ bandwidth-one LP. Two independent computations of that LP (primal and dual, lattice sizes to 64, mesh to 1/80) converge to ≈ 0.679–0.680 (Aitken 0.67940); this is the numerical value of a discretised LP with extrapolation, not a proved sharp constant, since the continuum limit carries no rigorous error control. Triple-correlation (Rudnick–Sarnak band) constraints raise the small-lattice ceiling by 3–7%, shrinking with resolution. No correlation-only construction beats the mollifier value 19/27; the M_{-} lemma above is the only identified in-method route past 0.6725.

7. An explicit refutation criterion for the alternative hypothesis (Astra) [P under RH]

For fixed $c > 0$ let $I_T(c) = \int_0^T |\zeta'/\zeta(\tfrac{1}{2} + c/\log T + it)|^{2}\, dt$ and

$W_T = 2[\sinh(2)\cdot I_T(1) - \sinh(1)\cdot I_T(\tfrac{1}{2})] / (T \log^{2} T)$.

Theorem 7.1. Assume RH and AH-pairs (the precise pair-correlation form of the alternative hypothesis in Lagarias–Rodgers). Then $W_T \to W_{AH}$ with $0.06239 < W_{AH} < 0.06240$ (exact rational enclosure). The sine-kernel prediction is 0.0822714431…. Hence a proof, under RH, that $\liminf_{T\to\infty} W_T \ge 1/16 = 0.0625$ refutes AH-pairs; the certified margin is $1/16 - W_{AH} > 0.00010$.

Theorem 7.2 (short-prime projection). Under RH the mean square in 7.1 equals its short-prime diagonal ($N = \lfloor T/\log^{6} T \rfloor$) plus a residual norm squared plus $O_c(N \log^{4} T)$; the residual has an absolutely convergent centred-ψ continuation whose pole can be removed at cost $O(\log^{-3} T)$; the short-prime two-scale main term is $B \in (0.45609397932923, 0.45609397932924)$; so the sufficient target is a lower bound $\liminf E_T \ge 1/16 - B$ on the residual energy. Here E_T is a signed difference of two residual energies, $E_T = (2/(T \log^{2} T))[\sinh(2)\|R_1\|^{2} - \sinh(1)\|R_{1/2}\|^{2}]$; the negative term cannot be discarded, and positivity of residual norms does not by itself deliver the inequality.

Twenty further reductions (complementary-modulus Möbius–log correlations to $X^{0.523}$, an RH component bound $X^{1.023}\log^{5} X$, exact Type-I removal below $X^{0.477}$, a full Vaughan reduction to a signed remainder, a full actual-variance reduction with its singular-series constant, removal of a central divisor band) narrow the missing estimate to a single signed prime correlation at scale $X \log X$. None proves it. What is new here is not a theorem about zeta but the exact location and size of the obstruction: an inequality with a five-decimal target that any future arithmetic method must clear.

8. Smaller exact results worth recording

- Residual-Gram saturation [P, Astra]. For a fixed resonator r and product cutoff L, the approximator-dependent main term in Inoue's half-gap method satisfies $2\mathrm{Re}\langle b_a, b_g\rangle_L - \|b_a\|^2_L = \|b_g\|^2_L - \|b_a - b_g\|^2_L$ with $b_a(q) = \Sigma_{km=q} a(k)r(m)$ and $\langle b,c\rangle_L = \Sigma_{q\le L} \bar{b}(q)c(q)/q$; so the coefficient-space approximator is exactly saturated on its own support and no rewriting on the same support recovers energy. Sparse longer tails under sub-polynomial coefficient bounds cannot supply constant energy.
- A mass-weighted Fock-space bound [P]. On the bosonic Fock space over $L^2((0,1], du/u)$ restricted to total mass ≤ 1, $\|a(g)\|^2 \le \int_0^1 |g(u)|^2/u^2\, du$; for the diagonal operator $K = A^*A + (A^2 + A^{*2})/2$ with $g(u) = 2\sin(\pi u/2)$ this gives $\|K\| \le 2(2\pi\cdot \mathrm{Si}(\pi) - 4) \approx 15.272$, and its truncated spectrum converges to $\lambda_\infty \approx 4.6456$, 6% below the threshold $\pi^2/2$, matching an unrelated symmetric-prime-feature search (4.6455) to four figures [C]. Whether the true arithmetic operator stays below $\pi^2/2$ is open.
- Complementary dense divisibility [C, Astra]. With $Y = 10$, $D = 330 = 2\cdot3\cdot5\cdot11$, $E = 455 = 5\cdot7\cdot13$, $f(p) = p$, $g(p) = p^2$, $A_0 = 121$, $C_0 = 2197$, $X = 27000$: the merged modulus $Q = [D,E] = 30030$ is triply 10-densely divisible although E alone is not — an exact witness that the rootwise predicates of the 186 mechanism genuinely exceed per-root dense divisibility, with shared primes allowed.
- The single-dislocation constant $s^* = 1.419640342\ldots$ (§3.2) and the threshold gap $\lambda^* = 0.4719538$ (§3.6), both exact to the stated digits and obtained by two independent routes each.

9. Status summary

result	tag	novelty check (6 Sept 2026)
Layer-cake theorem; H_2, H_3, H_4 records; BE-free certificates	P/C	no $m \ge 2$ improvement found in 2023–2026 literature
Signed-sieve no-gain	P	no treatment of debt-charged signed MT weights found
Theorem A, Theorem C(i), frozen moments, generator theorem, force energy	P	no published collision-time law for the finite flow found
Theorem B′, CUE stiffness tail, CUE depth law (two routes)	P modulo cited gap law (+ window stability on route i)	isolation-lemma novelty audit incomplete (Lehmer-pair kinship)
CβE law $N^{-2-2/(\beta+1)}$	P modulo (U3), Feng–Wei	—
LR exchange rate; λ^*	P / C	Lagarias–Rodgers do not consider dynamics
Fibre dimensions; centre-of-mass family; Theorem E1	C / P	Rodgers–Vallabhaneni compute moments, not the fibre
Nyquist conservation law; $\Phi(\rho)$; fourth-moment covariance; holonomy TV law	P / P/C	no occurrence of $\Phi(\rho)$ or the covariance formula found
Galerkin ladder; N_d decoding; flat frontier; edge no-go; weight freezing; rigidity; Φ_3	P	statements about a 2026 theorem; no prior analysis found
Pair-correlation LP ≈ 0.679 (numerical)	C	corrects a number in the programme's own files
3-block limit = 2 with full expansion; midpoint-insertion principle; discriminant–force identity	P	no prior collision-time constants for lattice defects found; the identity is elementary and was not separately searched
Ceiling conjecture false at N = 12; domain-wall constant $C^* = 2.11466488$; corrected conjecture	C / O	single-proposer file, constants cross-checked
Last-mile barrier theorem; $W_T = \int K F_T$	P under RH	restates Tao's remark quantitatively; GGÖS floor is GRH
W_{AH} criterion; short-prime constant B	P under RH	no prior explicit AH-pairs target found

Not claimed anywhere in this paper: RH, Montgomery's conjecture, $\mu = ½$, a refutation of AH, or a prime gap below 186.

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Source files (branch `claude/riemann-zeta-random-matrix-udxp3f`, `research/riemann-rmt/`): `H2_H3_record_announcement.md`, `overnight/fable/r1_h2_interval_cert.md`, `signed_sieve_nogo.md`, `depth_scaling_theorem.md`, `overnight/fable/r1_theoremB_repair.md`, `r1_cue_background.md`, `r1_cbe_background.md`, `r1_levelB_barrier.md`, `r1_structure_review.md`, `r1_simple_zeros.md`, `r2_diagonal_operator_spectrum.md`, `impostors_paper.md`, `final_verified_paper.md`, `round3_synthesis.md`; and in `QingyunSun/Riemann-hypothesis-and-random-matrix` (`research/reports/`): `yau_flow.md`, `yau_flow_galilean_refinement.md`, `dynamic_generator.md`, `force_energy.md`, `residual_gram_round1.md`, `dyson_round7.md`, `dyson_round8.md`, `dyson_round14.md`, `prime186_structural_frontier.md`, and `research/claims/CLAIM_LEDGER.md`.